

# Chankyu (Charlie) Han

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## RESEARCH INTERESTS

Smart Sensing and Haptic with AI/ML, Tactile Sensing, Human-Computer Interaction, Fabrication

## EDUCATION

### University of Washington (UW)

Seattle, Washington

Ph.D. in Electrical and Computer Engineering, Advisor: Dr. Yiyue Luo

Jun.2025 - Present

### Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, Korea

M.S. in Mechanical Engineering, Advisor: Dr. Inkyu Park

Mar.2021 - Feb.2023

B.S. in Mechanical Engineering  
Graduation with *Cum Laude*

Sep.2016 - Feb.2021

## PUBLICATIONS

(<sup>†</sup>: equal contribution)

### [1] Thermoforming 2D Films into 3D Electronics for Customizable Tactile Sensing

J. Choi<sup>†</sup>, **C. Han<sup>†</sup>**, D. Lee<sup>†</sup>, H. Kim, G. Lee, J.-H. Ha, Y. Jeong, J. Ahn, H. Park, H. Han, S. Cho, J. Gu, I. Park

*Science Advances* 11.20 (2025): eadv0057. <https://doi.org/10.1126/sciadv.adv0057>

### [2] Spike-based Self-calibration for Enhanced Accuracy in Self-powered Pressure Sensing

**C. Han<sup>†</sup>**, J. Choi<sup>†</sup>, J. Ahn, H. Kim, J.H. Ha, H. Han, S. Cho, Y. Jeong, J. Gu, I. Park

*Advanced Materials Technologies* (2023): 2301199. <https://doi.org/10.1002/admt.202301199>

### [3] Recent Advances in Sensor-Actuator Hybrid Soft Systems: Core Advantages, Intelligent Applications, and Future Perspectives

**C. Han<sup>†</sup>**, Y. Jeong<sup>†</sup>, J. Ahn<sup>†</sup>, T. Kim, J. Choi, J.-H. Ha, H. Kim, S.H. Hwang, S. Jeon, J. Ahn, J.T. Hong, J.J. Kim, J.H. Jeong, I. Park

*Advanced Science* (2023): 2302775. <https://doi.org/10.1002/advs.202302775>

### [4] Nanotransfer Printing of Functional Nanomaterials on Electrospun Fibers for Wearable Healthcare Applications

J.-H. Ha, J. Ko, J. Ahn, Y. Jeong, J. Ahn, S. Hwang, S. Jeon, D. Kim, S. A. Park, J. Gu, J. Choi, H. Han, **C. Han**, B. Kang, B.-H. Kang, S. Cho, Y.J. Kwon, C. Kim, S. Choi, G.-D. Sim, J.-H. Jeong, I. Park

*Advanced Functional Materials* (2024): 2401404. <https://doi.org/10.1002/adfm.202401404>

### [5] Wireless, Multimodal Sensors for Continuous Measurement of Pressure, Temperature, and Hydration of Patients in Wheelchair

S. Cho, H. Han, H. Park, S.-U. Lee, J.-H. Kim, S.W. Jeon, M. Wang, R. Avila, Z. Xie, K. Ko, M. Park, J. Lee, M. Choi, J.-S. Lee, W.G. Min, B.-J. Lee, S. Lee, J. Choi, J. Gu, J. Park, M.S. Kim, J. Ahn, O. Gul, **C. Han**, K. Lee, S. Kim, K. Kim, J. Kim, C.-M. Kang, J. Koo, S.S. Kwak, S. Kim, D.Y. Choi, S. Jeon, H.J. Sung, Y.B. Park, Y.T. Choi, M. Je, Y.S. Oh, I. Park

*npj Flexible Electronics* 7.1 (2023): 8. <https://doi.org/10.1038/s41528-023-00238-3>

- [6] [Battery-Free, Wireless, Ionic Liquid Sensor Arrays to Monitor Pressure and Temperature of Patients in Bed and Wheelchair](#)  
H. Han, Y.S. Oh, S. Cho, H. Park, S.-U. Lee, K. Ko, J.-M. Park, J. Choi, J.-H. Ha, **C. Han**, Z. Zhao, Z. Liu, Z. Xie, J.-S. Lee, W.G. Min, B.-J. Lee, J. Koo, D.Y. Choi, M. Je, J.-Y. Sun, I. Park  
*Small* 19.9 (2023): 2205048. <https://doi.org/10.1002/sml.202205048>
- [7] [All-Recyclable Triboelectric Nanogenerator for Sustainable Ocean Monitoring Systems](#)  
J. Ahn<sup>†</sup>, J.-S. Kim<sup>†</sup>, Y. Jeong, S. Hwang, H. Yoo, Y. Jeong, J. Gu, M. Mahato, J. Ko, S. Jeong, J.-H. Ha, H.-S. Seo, J. Choi, M. Kang, **C. Han**, Y. Cho, C. H. Lee, J.-H. Jeong, I.-K. Oh, I. Park  
*Advanced Energy Materials* (2022): 2201341. <https://doi.org/10.1002/aenm.202201341>
- [8] [A Review of Recent Advances in Electrically-Driven Polymer-based Flexible Actuators: Smart Materials, Structures, and Their Application](#)  
J. Ahn<sup>†</sup>, J. Gu<sup>†</sup>, J. Choi<sup>†</sup>, **C. Han**, Y. Jeong, J. Park, S. Cho, Y. Oh, J.-H. Jeong, M. Amjadi, I. Park  
*Advanced Materials Technologies* (2022): 2200041. <https://doi.org/10.1002/admt.202200041>
- [9] [Stretchable Printed Circuit Board Based on Leak-Free Liquid Metal Interconnection and Local Strain Control](#)  
M. S. Kim, S. Kim, J. Choi, S. Kim, **C. Han**, Y. Lee, Y. Jung, J. Park, S. Oh, B.-S. Bae, H. Lim, I. Park  
*ACS Applied Materials & Interfaces* 14.1 (2021): 1826-1837. <https://doi.org/10.1021/acsami.1c16177>
- [10] [Customizable, Conformal, and Stretchable 3D Electronics via Pre-distorted Pattern Generation and Thermoforming](#)  
J. Choi, **C. Han**, S. Cho, K. Kim, J. Ahn, D. D. Orbe, I. Cho, Z. Zhao, Y. Oh, H. Hong, S. S. Kim, I. Park  
*Science Advances* 7.42 (2021): eabj0694. <https://doi.org/10.1126/sciadv.abj0694>

## TECHNOLOGY TRANSFER

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Joint transfer of the following technologies (\$37,000, 2022):

Finger wearable touch sensor, control system using same, and method thereof

I. Park<sup>†</sup>, J. Choi<sup>†</sup>, **C. Han**<sup>†</sup>, G. Lee<sup>†</sup>

*Korea Patent (Application Number: 10-2021-0096964), 2021.*

Large-Area Multi-Pressure Sensor Pad Fabrication Technology (Know-how)

I. Park<sup>†</sup>, J. Choi<sup>†</sup>, **C. Han**<sup>†</sup>, G. Lee<sup>†</sup>

## RESEARCH EXPERIENCES

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### UW Wearable Intelligence Lab

Research Assistant

Mentor: Yiyue Luo

Jun.2025 - Present

### Korea University Sensor and Nanofabrication System Lab

Visiting Researcher

Mentors: Junseong Ahn, Jungrak Choi

Jan.2025 – Jun.2025

### KAIST Machine Intelligence and Novel-sensor Technology Lab

Undergraduate Intern & Research Assistant

Jun.2020 – Jun.2023

Chankyu (Charlie) Han

Last update: Oct.2025

Mentor: Inkyu Park

## SCHOLARSHIPS & AWARDS

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Best Oral Presentation Award</b><br>IEEE NEMS 2023 [2]                                                                                                                | May 2023               |
| <b>Best Poster Award</b><br>ICAE 2021 [10]                                                                                                                               | Nov. 2021              |
| <b>National Excellence Scholarship (Graduate Program)</b><br>Government funding for admission fees, full amount of school support fees in the master's program           | Mar.2021 – Feb.2023    |
| <b>KAIST Undergraduate Research Grant</b><br>Title: Customizable, Conformal, and Stretchable 3D Electronics via Pre-distorted Pattern Generation and Thermoforming [10]  | Sept. 2020 - Feb. 2021 |
| <b>Dean's List, School of Freshman</b><br>Top 2% of KAIST freshman cohort                                                                                                | Feb. 2017              |
| <b>National Excellence Scholarship (Undergraduate Program)</b><br>Government funding for admission fees, full amount of school support fees in the undergraduate program | Sep.2016 – Feb.2020    |

## SKILLS

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**Languages:** Python, C/C++, MATLAB, LabVIEW

**Fabrication:** 3D printing (DLP, FDM), Thermoforming, Laser processing(CO<sub>2</sub>, UV, Fiber), PCB design

**TOEFL:** 108/120

## OTHERS

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|--------------------------------------------|---------------------|
| Military Service in Republic of Korea Army | Jun.2023 – Dec.2024 |
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